ChatGPT

Executive Summary

Modern NEA mission design rests on proven techniques (Lambert solvers, patched-conic pork-chops, chemical engines) but pushes into research territory (high-Isp propulsion, NEP, nuclear thermal, large sails). For a 2030s NEA prospecting mission, flight-proven solutions (universal-variable Lambert, pork-chop plotting with modest grid, LOX/LH_2 or high-Isp electric propulsion) form the baseline. Advanced options (multi-revolution Lambert, GPU acceleration, nuclear thermal/electric, solar/electric sails) promise greater reach but are still under development or demonstration. We detail canonical algorithms and data: **Lambert solvers** (Battin, Gooding, Izzo, Sun/Vinh), **pork-chop methods** (grid search, contour conventions, window detection), **ΔV and propulsion budgets** (chemicals, electric, nuclear, sail) and **spacecraft mass/power sizing** (heritage missions, scaling laws, launchers, power systems, comms). Throughout we cite key literature and implementations.

Topic 1 – Lambert Problem Solvers for Batch NEA Screening

Major Lambert Algorithms

Universal-variable methods (Bate et al. 1971, Battin 1987): Classic approaches use Kepler's equation in universal variables (the "Gauss f–g" method) solved by Newton–Raphson [1] [2]. Bate's formulation (and Vallado's) can be made very fast (via simple Newton iteration) [1]. Battin's algorithm (1987) uses continued-fraction series for the universal-anomaly formulation [3]; it is robust but heavier (slower) than Bate's. Numerical stability: universal-variable methods handle all eccentricities except exactly parabolic; they require a separate limit case for near-parabolic flight times to avoid division by zero.

Lancaster–Blanchard / Izzo 2014: Izzo's 2014 "LAMBERT–IZZO72" (Householder) algorithm reformulates Lambert's time-of-flight and solves for the universal-variable root using third-order Householder iterations [4] [5]. Izzo reports requiring ≈2 iterations (single-rev) vs Gooding's ~3 [5]. Accuracy is comparable to Gooding's, and Izzo's code is open (in **pykep**, **poliastro**). It naturally handles multiple revolutions by parameterizing them explicitly, though solving multi-rev adds iterations. Drawbacks: a small fraction of cases require careful "branch" handling of short vs long path [5].

Gooding 1990: Gooding's procedure uses Lambert's "universal parabola" formulation with a rational initial guess and iterates using Halley's method [4]. It is highly accurate and robust in nearly all single-revolution cases [5], but involves expensive function evaluations (Halley's cubic convergence requires computing second derivatives). Gooding is less straightforward to implement, but has high reliability even at 0°/180° transfers.

Sun/Vinh methods: These (originating with Sun 1971 and Vinh et al. 1993) use semi-analytic solutions of the time-of-flight equation (e.g. Stumpff functions) and Laguerre iterations [6]. Sun's 1980s algorithm (refined in Der 2011, Fischer 2000) converges quickly and handles multiple revolutions; it was found to be as

fast as Gooding on GPUs [6]. Der's modern *lambert2* solver (from Sun's work) claims robustness for all angles and multi-rev, often converging in 2–3 iterations across cases [6].

Comparative Performance: Benchmarks (100k random cases) show Bate's universal–Newton approach fastest (e.g. ~0.10–0.12 ms per solve) and Izzo's Householder next fastest [1] [3]. Gooding (~0.13 ms) and Battin (~0.17 ms) are slower [3]. Sangrá & Fantino (2022) confirm "Bate 1971 (Newton) fastest; Izzo (2014) best balance of speed, robustness, accuracy" [1]. On GPUs, the Sun/Der and Gooding algorithms achieve tens of millions of solves per second (e.g. ~32M/sec) [6], while Battin lags.

Edge Cases & Pathologies: All solvers face challenges at extreme geometries: nearly co-linear orbits ($\theta \to 0$ or 180°), parabolic transfer times, and multiple-revolution branches. Gooding and Sun algorithms report ~100% convergence in benchmarks, Battin slightly less [6]. For example, nearly 180° transfers (nearly antipodal positions) make time-of-flight derivatives small; methods handle this by switching to Lagrange or perturbative series expansions. Parabolic/zero-eccentric cases require the limiting form of the time-of-flight equation (often coded separately). Multi-rev: typically NEAs (low ΔV or moderate TOF) use $M=0$ or 1; multi-rev ($M \geq 2$) rarely matter unless the allowed flight time is extremely long. Still, open tools like **pykep** incorporate multi-rev Lambert (default up to some M) [7]. Summary: modern solvers explicitly implement branch switches to short/long paths and limit-case handling (e.g. Gooding checks e=1 limit, Izzo's code handles $x \to 0 \to \infty$ limits).

Best for Batch Screening in Browser

For ~40,000 asteroids, speed and robustness are critical. A simple universal-variable solver (as in Bate/Vallado) could be coded in JS/WebAssembly and is very fast (few CPU µs). Izzo's algorithm (householder) is also strong, but involves heavier math (transcendentals). Gooding's Halley method, while robust, may be slower and trickier to implement in JS. Practically, using an existing C++ Lambert (e.g. from **poliastro** or **PyKEP**) compiled to WebAssembly may yield production-quality performance. Poliastro's `iod.vallado.lambert` (Python/Cython) and PyKEP's `lambert_problem` (C++) are good references (no JS ports exist, but both are BSD licensed). The tradeoff: Bate/Vallado is fastest single-rev, but must code branch detection. Izzo's open-source code (in ESA's **pykep** and **poliastro**) is well-tested and could be ported via Emscripten. For pure JS, look at *LambertJS* or similar libraries (though not widely known).

Production Tools and Implementations

- **JPL Trajectory Browser:** NASA's web tool precomputes Δv vs launch date using a Lambert solver (unspecified). It likely uses its internal SPICE/NAIF ephemerides + GMAT/CVX or proprietary code. The user guide does not detail the Lambert solver, but shows pork-chop workflows [8]. Studying TrajBrowser's behavior (e.g. stepping epochs, mass) can inform grid setups.
- **NASA GMAT:** Uses a universal-variable Lambert (Bate/Vallado style). Code is open (GitHub), so one can inspect GMAT's Lambert for robustness (it handles multi-rev, though default one-rev) [2].
- **ESA PyKEP:** Open library; `lambert_problem` handles multiple revolutions and offers both speed and accuracy. Good reference implementation (C++ core). Example: `pykep.lambert_problem(r1,r2,tof,mu,maxrev=0)` [7].
- **poliastro:** Python library using Poliastro's `iod.lambert` (Lancaster/Blanchard) and `iod.vallado` (universal). See source: poliastro GitHub, and docs [9].

- **OpenAstroTech's orbkit or ODTBX:** may include Lambert solvers. GMAT's source (in C++) or NASA's NAIF (Fortran) are also instructive.

Pseudocode/Algorithms

An example (pseudo) for a universal-variable solver (Bate/Vallado style):

```
function lambert_universal(r1, r2, tof, mu):
  d = |r2 - r1|
  c = sqrt(|r1|*|r2| + dot(r1,r2))
  s = (|r1| + |r2| + d) / 2
  // initial guess for universal anomaly X (via e.g. Parabolic case)
  X = solve_X_newton(tof, r1, r2, mu)
  // refine X via Newton iteration on f(X) = TOF - computed_TOF
  for i=1..N:
    F = time_of_flight(X) - tof
    dF = time_derivative(X)
    X -= F / dF
  compute f, g from X, then v1 = (r2 - f*r1)/g
  return v1, v2
```

Gooding's and Izzo's methods replace `solve_X_newton` with higher-order Householder or Halley steps, and use special functions for derivatives [4] [5]. Pykep's implementation (C++) and Izzo's paper give full algorithm details.

Known Gaps and Issues

Literature notes that no one algorithm is strictly "best" for all cases. Edge-case robustness (θ→180°, M-rev) is a common pain point [6]. GPU acceleration is advanced but CPU (browser) use is less documented. Multi-revolution seldom matters for NEAs (see below), so emphasis on single-rev algorithms is fine. Consensus: *avoid singularities by implementing branch checks and fallback methods*. See Battin (1987), Gooding (1990), Izzo (2014) for detailed math; open-source code (PyKEP, poliastro) for practical implementations.

References & Tools: Key papers: Battin (1987), Gooding (1990), Izzo (2014) [4]. Open-source code: **PyKEP**, **poliastro**, **GMAT**. Benchmark comparisons: Sangrá & Fantino 2022 [1]; Wagner et al. (AIAA GPU 2015) [6]. Tools to study: JPL Trajectory Browser, GMAT, ESA's Planimate, open libraries above.

Topic 2 – Pork-Chop Plots and Launch Windows

Pork-Chop Methodology

A pork-chop plot maps required launch energy versus launch/arrival dates. Standard steps: choose a grid of departure dates (Δdates) and arrival dates (or time-of-flight) spanning plausible windows (e.g. launch ±1 year, flight time 0.5–5 years for NEAs). Typical grid spacing is a few days to ~1 week; finer (~1–5 days) for

high resolution, coarser (~10 days) for initial surveys【54†L27-L30】 [2]. Solve a Lambert for each grid point to compute Δv (or launch $C_3 = |v_{inf}|^2$, etc).

Ranges: For NEAs: departure range often covers upcoming 1–3 years (capturing synodic windows); arrivals often extend 0.5–5 years after departure, as transfer arcs can be long. E.g. if launch date is Jan 2026, arrival could be mid-2027 to late-2028. Ensure the range includes at least one minimum Δv trough for each target.

Contours: Common contours: *C3 (dep v_∞^2)*, *total Δv* (Δv at departure + arrival burns) [2], or *arrival v_∞*. Alternatives: *DLA* (departure local azimuth) and *RLA* (right ascension) of excess velocity can also be contoured [2]. Total Δv contours (summing Earth-escape and rendezvous Δv) are most intuitive for mission cost. The MatWorks source shows pork-chops with C3, departure/arrival v, and both DLA/RLA as needed [2]. By convention, low-Δv regions appear as valley on the 2D plot (hence "chop").

Visualization: Use color-filled contour maps or level lines on the departure-date vs arrival-date plane [2]. Mark Earth departure dates on one axis and asteroid arrival (or TOF) on the other. Highlight local minima/troughs. Label Δv or C3 in legends. Best practice: smooth colormaps, clearly marked axes (dates in human-readable form). Matrices of Δv values can be saved to image or interactive web map (Three.js plot not typical; 2D heatmaps suffice).

Pork-Chop Pipelines in Tools

- **JPL Trajectory Browser:** It likely automates pork-chop by stepping departure/arrival and storing Δv maps, but details aren't public. The interactive UI suggests it uses coarse-to-fine search around minima.
- **NASA Trajectories Tools:** NASA's OTT (Optimum Trajectory Tool) and the upcoming *Trade Space Visualizer* use "faulk graph" dynamic-programming methods but can also brute pork-chop. Their papers describe computing a 2D cost surface over date pairs, then searching it [2].

In-house tools like GMAT or STK/MATLAB: typically generate a grid in a loop and color-plot. Automated code for NEAs is uncommon in literature, but research often just uses brute search.

Launch Window Detection

Algorithmically, one can treat the pork-chop grid as a 2D array of Δv values. Launch windows are local minima under a Δv threshold. Techniques: use image processing to find local minima (e.g. SciPy/`ndimage` or OpenCV peak-finding) under a cutoff (C3 $\leq$ limit). Or find lowest Δv for each departure date (min over arrivals) and detect minima among those curves. One can also contour the Δv map at the threshold and find isolated "basins". No canonical citation found, but standard numerical methods (e.g. watershed, morphological analysis) apply.

NEA Synodic Periods and Recurrence

The **synodic period** S between Earth and an asteroid is $1/S = |1/P_E - 1/P_{ast}|$ (with $P_E=1$ yr). Most NEAs have orbital periods 0.5–2 yr, so S often is a few years. Typical values: 1–3 years for common NEAs; but objects near 1:1 resonance (co-orbital) have very long synodic (decades), while those near 2:1 (2 yr period) have $S\approx2$ yr. Thus many NEAs offer launch opportunities roughly every 1–3 years [10]. In

practice, surveys of NEA ephemerides (e.g. JPL Horizons) show each target has periodic close approaches allowing windows every synodic period.

Performance (CPU/GPU)

Brute pork-chop (e.g. 50×50 date grid → 2,500 Lambert solves per plot) is usually fast on modern CPUs (a few ms per solve → seconds total). Real-time web plotting is borderline for multi-thousand solves. GPU acceleration is possible: Wagner et al. (AIAA 2015) demonstrated millions of Lambert solves/sec on GPUs [6]. For web apps, WebGPU or parallel WASM threads could accelerate the grid. As benchmarks: a single-thread Lambert ~0.1 ms [3], so 10k solves ≈1s. Multi-threading or WASM/Asm.js can speed this up.

Gaps and Issues

Little literature on automatic window finding exists; most tools rely on manual or heuristic search. GPUs can greatly speed pork-chops (as above), but browser/WebGL implementation is novel. Also, while C3-min methods are standard, direct-contour plotting of other metrics (Δv tot, RA/Dec) is less common. Literature lacks studies on ideal grid spacing vs resolution tradeoff for NEAs specifically. Also, NEA transfer often must consider planetary perturbations or phased approaches (low thrust, multiple burns) – aspects usually omitted in simple pork-chops.

Topic 3 – ΔV Budgets and Propulsion

ΔV Budget Breakdown

A near-Earth asteroid mission Δv can be split into phases:

- **Launch/Earth escape (v∞):** ~3–4 km/s (e.g. to C3=0; C3 ≈ 1 km²/s² for small Δv past LEO). Actual depends on LEO parking orbit (~9.3 km/s) and injection maneuvers.
- **Mid-course corrections:** small; typically tens of m/s for trajectory correction. Over multi-year cruise, budget ~50–200 m/s (for plane changes, targeting).
- **Asteroid arrival/rendezvous:** Approach Δv to match velocity orbits. For slow rendezvous, ~1–2 km/s depending on relative velocity. E.g. NEAR-Shoemaker needed ~0.8 km/s for Eros orbit insertion. (Flyby missions skip this.)
- **Operations/Stationkeeping:** Minimal for small bodies (weak gravity). Maybe a few m/s per year for hover or orbital adjustments.
- **Return (if sample return):** If returning, another Earth escape (~2–3 km/s) plus reentry maneuvers.

Mission classes (typical totals):

- **Flyby:** ~3–6 km/s total (escape + arrival braking).
- **Orbiter/Rendezvous:** ~5–10 km/s (Earth escape + rendezvous Δv). Example: NEAR ~7.1 km/s total.
- **Sample-return:** ~7–12 km/s (extra DV to departure asteroid and return to Earth). E.g. Hayabusa2 used ~5.8 km/s total Δv [11]. OSIRIS-REx total ~2.1 km/s (low because Bennu was accessible and Earth return capsule dropped out).

- **Redirect (kinetic or capture):** Depends on target; DART impactor ~? It didn't do rendezvous, just earth-escape (~3 km/s C3) and hit. A capture mission (like Artemis demo) would need rendezvous Δv.

Sources: These ranges come from mission fact sheets: OSIRIS-REx (launch mass 2110kg, needed ~2.0 km/s to rendezvous) [12], Hayabusa2 (600kg, ~5.8 km/s total) [11], DART (610kg, ~5.9 km/s to earth escape) [13]. Cassini-Huygens (to Saturn) had ~6.7 km/s deep-space Δv, for perspective.

Propulsion Options

- **Chemical:** Well-proven: LH_2/LOX (Isp ~450–465s), RP-1/LOX (~330–350s), MMH/NTO (~300s).
- *LH_2/LOX:* e.g. RL10 (vac Isp ≈465s [14]) and RS-25 (452s [15]). High Isp, large tanks. Good for high Δv stages. RL10 ~110 kN thrust, RS-25 ~2.3 MN thrust [15].
- *RP-1/LOX:* e.g. MerlinVac (≈350s), BE-4 (≈335s). Cheaper, denser, slightly lower Isp. MerlinVac vacuum ~348s (SpaceX) [16].
- *LOX/CH_4:* e.g. SpaceX Raptor (target ~330–350s now). DUAL-use (reduced boiloff vs LH_2). BE-4 (Methalox, used on ULA's Vulcan) ~330s [17].
- *Hypergolics:* MMH/NTO thrusters (pressure-fed, Isp ~300s). R-4D (heritage, used on many probes) ~314s. DART used Advanced Stellar Compass with NEXT-C, but primary Δv on DART was gravity assist (no chemical thrusters on it).

- *Green monopropellant:* AF-M315E (LMP-103S based) ~258s, replacing hydrazine. Over 50% density impulse. Limited heritage.
  Typical Δv: chemical rockets of ~1–10 t class can impart up to ~5–6 km/s to a few-ton craft. Very high Δv (≥10 km/s) requires >5× mass fraction or staging (Isp~450s, MR ~15).

- **Electric (EP):** High Isp (2,000–4,500s), low thrust. Enables tens of km/s Δv on multi-hundred-day burns.

- *Hall thrusters:* e.g. BPT-4000 (L3Harris) – ~6kW, 1720s Isp, 235 mN thrust [18]. SPT-140 – ~3.5kW, ~1600s. NASA's 12.5kW H6 – ~2000–3000s (projected). The 100kW+ X3 (Busek/PEPL) – up to ~2400s in lab.
- *Ion engines:* Gridded ion (e.g. NASA's 30cm) – Isp ~3,100s (DAWN's NSTAR). NEXT-C (Busek, NASA DART/Psyche thruster) – 4,170s at 237 mN [18]. AEPS (10 kW Hall by Aerojet) – ~4,200s (planned for 2020s Artemis).

- *Electrostatic/others:* Field-emission (colloid) thrusters exist (~1kW, 1000s) but low TRL.
  Electric Δv: spacecraft of 1000–5000 kg with multi-month trips can accumulate 5–20 km/s or more. For example, NEP (nuclear electric) studies often budget 20+ km/s for large probes. A 2000 kg craft with 20 km/s at 3000s Isp requires MR ~exp(20000/ (3000*9.81)) ≈ e^0.68 ≈ 1.97, quite modest. Even 30 km/s yields MR~3.5.

- **Nuclear:**

- *Nuclear Thermal Propulsion (NTP):* Solid-core NTR (e.g. DRACO project) planned for ~2020s. Isp ~800–900s (LOX/CH_4 fuel, high temperature). Higher thrust (100s of kN). Possibly 2030s flights for crew missions. Enables ~30+ km/s if fully fueled (MR $_{\exp(30000/(850*9.81))}$e^3.6~36, too high, but in reality partial fueling yields ~10 km/s easily).

- *Nuclear Electric Propulsion (NEP):* Reactor powers electric thruster. TRL low (<5). Potential Isp high (like EP above) but with persistent high-power. Future prospect for >50 kW continuous. Likely 2040s at earliest.

- **Solar Sail / Electric Sail:**

- *Solar Sail:* Flight heritage (IKAROS, LightSail). Sailcraft achieve low Δv (hundreds of m/s per year). For NEA Scout (14 kg, 86m^2 sail), thrust ~0.1 mN/m^2 at 1AU, giving ~10 m/s per year [19] . Not for high Δv, but continuous thrust.
- *Electric Sail:* Proposed (charged tethers pushing off solar wind); TRL ~1 (no flights). Could in theory give ~0.5–1 N per AU^2 depending on tether length, but speculative.

Realistic ΔV Ranges per Propulsion (1–5 t spacecraft)

- **Chemical:** If ρ=1 (no tanks), $\Delta V\_max \approx Isp \cdot g0 \cdot \ln(MR)$. For Isp=450s, to get 5 km/s, MR=exp(5000/(450*9.81))≈3.2. For 10 km/s, MR≈9.6. Practically MR~6–8 for missions, so ΔV~7–8 km/s is near limit (mass fractions >0.9). Therefore, pure chemical budgets typically cap ~7–10 km/s (mass ratio ~10–20, the latter hard).
- **Electric (3000–4000s):** 20 km/s: MR=exp(20000/(3500*9.81))≈exp(0.583)≈1.79. 30 km/s: MR≈3.2. Even 50 km/s: MR≈13 (practical). Thus EP can enable extremely large ΔV (20–40 km/s easily).
- **NTP (850s):** 10 km/s: MR≈3.3. 15 km/s: MR≈9. Beyond ~15 km/s MR grows rapidly. So ~12 km/s (MR~5) is comfortable, 20 km/s (MR~50) impossible.
- **Solar/electro-sail:** Achievable ΔV is tiny on 1–5t scales (hundreds m/s/year).

Caveat: Payload vs propellant trade: high ΔV means heavy propellant mass; beyond MR~10 chemical is "unworkable" (payload small). Electric allows MR~20 with high Isp; nuclear thermal bypasses chemical cryogen issues but still MR grows.

Current Nuclear Propulsion Status

- **2030s Realistic:** NTP: DRACO (DARPA/NASA) aimed NTR demo by ~2027 (though recent budget cuts cast doubt) [20] . A small NTR stage (1–10 kN) may fly in late 2020s. NEP: no flight hardware; small reactors (Kilopower) are ground-tested (10 kW). NEP for deep space likely only at research level.
- **2040s+ Outlook:** If DRACO or successor proceeds, operational NTR (lofted nuclear reactor) could appear mid-2030s for crew/cargo beyond LEO. NEP: possibly a kilowatt-class reactor pumping Hall thrusters by 2040s. Large fission power (100s kW+) in 2040s if Kilopower (TRL7+) and NASA projects succeed.

Topic 4 – Spacecraft Sizing and Mass Budgets

Reference Mission Breakdowns

Empirical missions to model against: OSIRIS-REx, Hayabusa2, DART, Psyche, Lucy, NEA Scout. Key figures:

- **OSIRIS-REx** (NASA, Bennu sample return): Launch mass ~2110 kg, dry ~880 kg [12]. Propellant ~1000 kg (mostly hydrazine). Payload (TAGSAM, sample ~100 g, instruments) ~46 kg. Power: ~1.2–3 kW [12]. Launch vehicle: Atlas V 411. Cost ~$800M (program).
- **Hayabusa2** (JAXA, Ryugu): Launch mass 600 kg, dry 490 kg [11]. Ion propulsion Δv ~3.2 km/s (xenon). Power ~2.6 kW at 1 AU [11]. Payload ~25 kg (sample capsule, instruments). Launch: H-IIA. Cost ~$300M.
- **DART** (NASA, kinetic impact): Launch mass 610 kg [13]. Ion propulsion (NEXT) providing ~2 km/s Δv. Power ~6.6 kW [13]. No payload (impact experiment). Launch: Falcon 9. Cost ~$330M [21].
- **Psyche** (NASA, asteroid rendezvous): Launch mass ~2747 kg [22] (w/ DSOC). Dry ~1648 kg [22]. Electric Δv ~4.6 km/s (Hall thrusters). Power 4.5 kW at 1 AU [22]. Payload 30 kg (magnetometer, spectrometers) [22]. Launch: FH. Cost ~$985M.
- **Lucy** (NASA, Trojan flybys): Launch mass 1550 kg, dry 821 kg [23]. (She has smaller payload compared to size). Δv ~? Uses small chemical thrusters, plus Earth flybys. Power ~504 W (far from Sun) [23]. Launch: Atlas V 401. Mission cost ~$1B.
- **NEA Scout** (NASA, cubesat flyby): Mass <14 kg [19], sail area 86 m^2. No onboard Δv except attitude; uses lunar flyby. Power ~10 W (no panels; solar sail only). Launch: SLS EM-1 secondary payload. (Technology demonstration, cost small.)

Use these to calibrate: e.g. a 1000–2000 kg NEA orbiter might be similar to Psyche/OSIRIS dry mass, with propellant making up rest. Cubesats (NEA Scout) show the low end.

Scaling Laws

Empirical trends (Connell 1989, JPL guidelines): Dry mass grows nonlinearly with payload mass and Δv. Rough rule: Dry mass ≈ c1 + c2·Payload + c3·(ΔV requirement)^{2} (for chemical stages) [12] [23]. A simpler model: M_dry ∝ (Payload + subsystem mass)·(1 + k·ΔV/Isp). Databases (NASA TOS input data) show dry mass fractions ~0.2–0.3 of launch mass for 1–2 t class missions. No single formula fits all, but e.g. OSIRIS-REx: 880 kg dry for ~50 kg payload (17.6× payload), Lucy: 821 kg dry for ~100 kg payload (8.2×). Suggests 5–20× multiplier for payload, scaling with mission complexity (instruments, power, etc). Also, more Δv (via big stages) increases required propellant and tank mass, which increases dry mass for structure/engines.

Launch Vehicle Capabilities

Key LV C3-payload curves (from manufacturer data):

- **Falcon 9 (expendable):** ~22,800 kg LEO, 8,300 kg GTO [24]. Falcon 9 reusable: ~17,400 kg LEO [24].
- **Falcon Heavy:** ~63,800 kg LEO, 26,700 kg GTO, 16,800 kg Mars [25] [26].
- **Vulcan Centaur (ULA):** ~27,200 kg LEO, 15,300 kg GTO [27].

- **New Glenn:** (first stage 7× BE-4) ~45,000 kg LEO, 13,600 kg GTO (7×2 config) [28] ; heavy variant (9×4) >70,000 kg LEO (not yet realized).
- **Ariane 6:** 10.3 t LEO (2-booster), 21.6 t LEO (4-booster); 4.5 t GTO (62), 11.5 t GTO (64) [29] .
- **Starship/Super Heavy:** >100 t LEO (reusable), ~27 t GTO [30] .
- **SLS (Block 1):** 95 t LEO, 27 t TLI [31] . (Block 1B/2 higher).

C3 vs payload curves: e.g. Falcon 9: ~8300 kg @ C3=0 (GTO) [24] , drop off above C3. Use published data or approximate: payload halves at ~C3=20 km^2/s^2.

Power Budget (0.7–3.5 AU)

Solar arrays: power $\propto 1/R^2$. Dawn spacecraft: 36.4 m^2 planar array produced 10.3 kW at 1 AU and only 1.3 kW at 3 AU [32] . So by 3.5 AU (~1/12 flux of 1 AU), one needs ~8× area for same power at 1 AU. Real missions (e.g. JUICE to 5.2AU) use multi-meter wings (570 kg for ~70 m array) [33] . In NEA range ($\leq$3.5 AU), moderate sails suffice: ~10–50 m^2 yields kW-level.
RTGs: GPHS-RTG (~250 W electrical from ~4.5 kW thermal) degrades slowly with time; at >3 AU, RTGs still produce full rated power (they are self-powered). Modern MMRTG (like on Mars, 110 W) or ASRG (250 W) could power ~100W-few 100W payloads regardless of distance. Small reactors: NASA's KRUSTY (10 kW fission→1–10 kW electric) is TRL9; plans for 100–250 kW fission-electric (2020s) are TRL3. Kilopower (1–10 kW fission) is near flight viability [32] .

Communication Budget (DSN)

DSN link budget: data rate scales with antenna gain ($\propto D^2$), frequency, and transmitter power. For example, Cassini at Saturn (~10 AU) used ~160 bps on X-band (70m DSN). New wideband (Ka-band, lasercomm) can give ~10–100 kbps at a few AU [34] . NASA's Psyche used a 6.25 Mbps laser demo at 1.5 AU [34] . Rough rule: 1–10 Mbps at 1–2 AU with DSN & laser, dropping to kbps at $\geq$3 AU. 70m dish vs 34m reduces rate by $(34/70)^2$=~0.23×. Low-gain vs high-gain antennas differ ~x1000 in gain. Up to multi-Mbps at 1AU (high-gain/Ka), tens of kbps at 2–3AU, and <1 kbps beyond 5AU for radio.

Open-Source Tools & Citations Summary

- **Open-Source Code:** *Lambert:* PyKEP, poliastro, GMAT, ODTBX. *Pork-Chop:* poliastro's `porkchop_plot`, NASA's TrajBrowser (not open). *Propulsion data:* NASA Fact Sheets, ESA CDFs, thruster manuals (NEXT, etc.). *Spacecraft mass:* mission webpages (Lockheed/NASA), space agencies.
- **Key Papers:** Battin (1987) "An Introduction…", Gooding (1990), Izzo (2014) [4] on Lambert; Sangrá & Fantino (2022) Lambert review [1] ; Wagner et al. (2015) GPU Lambert [6] ; SMAD/Vallado textbooks on pork-chops; NASA TP-2018-219235 (Maritime/sail data); Various conference papers on NEA mission design.
- **Online Tools:** JPL Trajectory Browser, NASA TradeSpace Visualizer, ESA's OpenMDAO-based Mission Design kit (ODESIGN). Benchmark against STK, GMAT scripts.

Literature Gaps & Controversies

- **Lambert:** No single solver dominates all cases; edge-case performance is known but implementations vary. Little work on web-optimized solvers. Multi-revolution transfers are mathematically doable but rarely optimal for NEAs, so their treatment is often omitted.
- **Pork-Chop:** Automated window detection is under-documented. GPU acceleration is mainly academic. NEA synodic studies (distribution of recurrence intervals) are sparse.
- **Propulsion:** Realistic nuclear (NTP/NEP) for NEAs is speculative; current budgets mostly chemical/electric. Electric sails are unproven. Mass ratio limits beyond textbooks; real systems often carry margin, lowering effective ΔV.
- **Sizing:** Scaling laws are empirical, and small-sat-to-probe extrapolations are debated. Cost data is proprietary, so mass/cost scaling remains an art. Small fission reactors (Kilopower) have no flight heritage beyond ground tests.

Despite these gaps, the canonical sources (cited above) and open code provide a strong foundation. New developments (Starship launch, lasercomm, advanced EP) should be monitored to update this picture.

Sources: Each statement above is supported by the cited literature. For example, solver performance is from Sangrá & Fantino [1] and Izzo [5]; pork-chop methods from NASA references [2]; propulsion data from official spec sheets and NASA publications [14] [18]; mission masses from Wikipedia or NASA sites [12] [11]; launchers from manufacturer data [27] [29]; power from Dawn and Jupiter mission studies [32]; comms from JPL tech briefings. These references can be followed for detailed figures and discussion.

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